



School of Future Tech

Case Study Report

on

Canvasback Terrarium & Houseplant Subscription Co.
Network Foundations, Topology, and IP Addressing Design

by

Ishita Nakhawa
150096725213

INDEX

Sr. No.	Title
1.	Introduction
2.	Problem Statement Background
3.	Problem Statement Design
4.	Methods & Algorithm/Technology Applied
5.	Implementation Details
6.	Snapshots
7.	Conclusion
8.	Reference

1. Introduction

A network is the road system of an organization: it lets computers, sensors, and printers talk to each other. This project designs and builds the network for Canvasback Terrarium & Houseplant Subscription Co., a business that is setting up a new greenhouse & fulfillment center and a remote propagation annex. Nothing exists yet, so the network has to be planned from scratch before any equipment is bought.

The design was planned on paper first (addressing plan, topology, and protocol mapping) and then built and tested in Cisco Packet Tracer. The project uses VLSM subnetting, VLANs, DHCP, DNS, ARP, and ICMP and maps everyday activities to the OSI and TCP/IP layers.

2. Problem Statement Background

The company is networking its main building, which has four working areas, plus a remote annex connected by a leased line. Management wants a written proposal before signing any purchase order.

Segment	Hosts needed	Notes
Greenhouse Growing Floor	36	Climate sensors, misting-system controllers
Packing & Fulfillment Area	16	Packing-line scanners, label printers
Administration	9	Admin laptops, a networked printer
IT	6	Servers, network devices
Propagation Annex WAN link	2	Point-to-point leased line

Objectives of the case study:

1. Recommend a topology and justify it by cost, fault tolerance, and ease of adding devices.
2. Map day-to-day activities to the OSI and TCP/IP layers (5 rows).
3. Divide 10.185.0.0/22 with VLSM, adding 20% spare capacity to every LAN.

4. Decide which addresses are public and which are private, and explain why the WAN link can use private addresses.
5. Produce a device list with the OSI layer each device works at.
6. Explain the order of DHCP, ARP, DNS, and ICMP for a new climate sensor.

3. Problem Statement Design

3.1 Topology recommendation: extended star

Every area connects to a central switch, and that switch connects to the router. The Annex connects to the router by a separate point-to-point link. Extra access switches can be added to the central switch later, which is what makes it an extended star.

Factor	Why extended star fits
Cost	Needs only short cables from each area to the switch. A mesh would need many more links and ports.
Fault tolerance	If one cable or device fails, only that device goes down; the rest keep working. The weak point is the central switch, which can be protected later with a second switch.
Ease of adding devices	A new device just plugs into a free switch port, or a new access switch is added. Nothing else needs to change.

Bus and ring topologies were not chosen because a single break stops the whole network. Full mesh was rejected because of its cost.

3.2 VLSM addressing plan (10.185.0.0/22)

Method: add 20% spare hosts, add 2 for the network and broadcast addresses, pick the smallest block that fits, and allocate the largest segment first so the blocks do not overlap.

Segment	Hosts	+20%	Block	CID R	Mask	Network	Usable range	Broadcast	Gateway
Greenhouse	36	44	64	/26	255.255.255.192	10.185.0.0	.1 to .62	10.185.0.63	10.185.0.1
Packing	16	20	32	/27	255.255.255.224	10.185.0.64	.65 to .94	10.185.0.95	10.185.0.65
Admin	9	11	16	/28	255.255.255.240	10.185.0.96	.97 to .110	10.185.0.111	10.185.0.97

IT	6	8	16	/28	255.255.255.240	10.185.0.112	.113 to .126	10.185.0.127	10.185.0.113
WAN link	2	2	4	/30	255.255.255.252	10.185.0.128	.129 to .130	10.185.0.131	n/a

Worked example: Greenhouse needs 36 hosts. $36 \times 1.2 = 43.2$, rounded up to 44. Adding 2 gives 46 addresses. The smallest power of 2 that is 46 or more is 64 (2^6), which leaves 6 host bits, so the mask is /26. IT needs a /28 and not a /29 because a /29 gives only 6 usable addresses, and IT needs 8 with a spare. The WAN link gets no spare because it only has two ends. Addresses from 10.185.0.132 onward are unused and available for future growth.

3.3 Public vs. private addresses

The whole design uses 10.185.0.0/22, which belongs to the private range 10.0.0.0/8 (RFC 1918). Private addresses are free, cannot be routed on the public internet, and are meant for use inside an organization. The WAN link to the Annex can safely use private addresses because it is a leased line owned by the company: traffic goes straight from one company router to the other and never crosses the public internet, so nobody outside needs to reach those addresses. If the company later needs internet access, an edge router would use NAT to translate private addresses to one public address.

4. Methods & Algorithm / Technology Applied

4.1 VLSM algorithm (steps)

1. List the segments and the hosts each one needs.
2. Add 20% spare capacity to each LAN and round up.
3. Add 2 addresses (network and broadcast).
4. Find the smallest power of 2 that fits, which gives the block size and CIDR.
5. Sort the segments from largest to smallest.
6. Allocate the blocks one after another starting at 10.185.0.0, each starting at the previous broadcast address plus 1.

4.2 Technologies used

Technology	What it does here
VLAN (IEEE 802.1Q)	Splits one switch into four separate departments (VLAN 10, 20, 30, 40), so broadcasts stay inside each department
Trunk port	One cable that carries all four VLANs between the switch and the router
Router-on-a-stick	One router port with four sub-interfaces (Gi0/0.10 to .40), each acting as the gateway for one VLAN

DHCP	The router hands out addresses, gateways and the DNS server to devices automatically
DNS	The IT server turns the name monitor.canvasback.local into 10.185.0.114
ARP	A device learns the MAC address of its gateway
ICMP (ping)	Tests whether a destination can be reached
Static default route	R-Annex sends everything it does not know to R-Main (10.185.0.129)

4.3 OSI / TCP-IP activity mapping

Everyday activity	OSI layer	TCP/IP layer
A climate sensor reports humidity to the greenhouse dashboard.	Layer 7, Application	Application
A firewall tracks an open subscription-system session.	Layer 4, Transport (it follows the connection state)	Transport
A router chooses the best path to the Propagation Annex.	Layer 3, Network	Internet
A switch forwards a frame using a MAC address table.	Layer 2, Data Link	Network Access
A faulty cable causes signal loss on the growing floor.	Layer 1, Physical	Network Access

5. Implementation Details

5.1 Devices used

Device	Model	Main OSI layer	Role
R-Main	Cisco 2911 router	Layer 3	Gateways for all four VLANs, DHCP server, and WAN link
R-Annex	Cisco 2911 router	Layer 3	Remote Annex end of the leased line
SW-Core	Cisco 2960-24TT switch	Layer 2	Connects all devices, carries the VLANs
GH-PC1, GH-PC2, GH-PC3	PCs (stand-ins for sensors)	Layers 1 to 7	Greenhouse hosts

PK-PC1, PK-PC2	PCs	Layers 1 to 7	Packing hosts
AD-LT1, AD-LT2	Laptops	Layers 1 to 7	Admin hosts
IT Server	Server	Layers 1 to 7	DNS server (static 10.185.0.114)
IT-PC1	PC	Layers 1 to 7	IT host
Cables	Copper straight-through and cross-over	Layer 1	Straight-through for PC and switch links and switch to router; cross-over between the two routers

No access points were used. The sample devices stand in for the full host counts (36, 16, 9, and 6), which the addressing plan already covers.

5.2 Build steps

1. Placed 2 routers, 1 switch, and 9 end devices and named them.
2. Connected 11 cables: switch to router (Gi0/1 to Gi0/0), router to router (cross-over), and the 9 end devices to switch ports Fa0/1 to Fa0/9.
3. On SW-Core: created VLANs 10 (GREENHOUSE), 20 (PACKING), 30 (ADMIN), and 40 (IT); assigned ports Fa0/1-3, Fa0/4-5, Fa0/6-7, and Fa0/8-9; and set Gi0/1 as a trunk.
4. On R-Main: enabled Gi0/0, created sub-interfaces .10, .20, .30, and .40 with dot1Q encapsulation and the four gateway addresses, set Gi0/1 to 10.185.0.129/30, and created four DHCP pools with the gateways excluded and .114 reserved for the server.
5. On R-Annex: set Gi0/0 to 10.185.0.130/30 and added the default route to 10.185.0.129.
6. Gave the IT server the static address 10.185.0.114 and added the DNS record monitor.canvasback.local to 10.185.0.114.
7. Set all other devices to DHCP.

5.3 Testing results

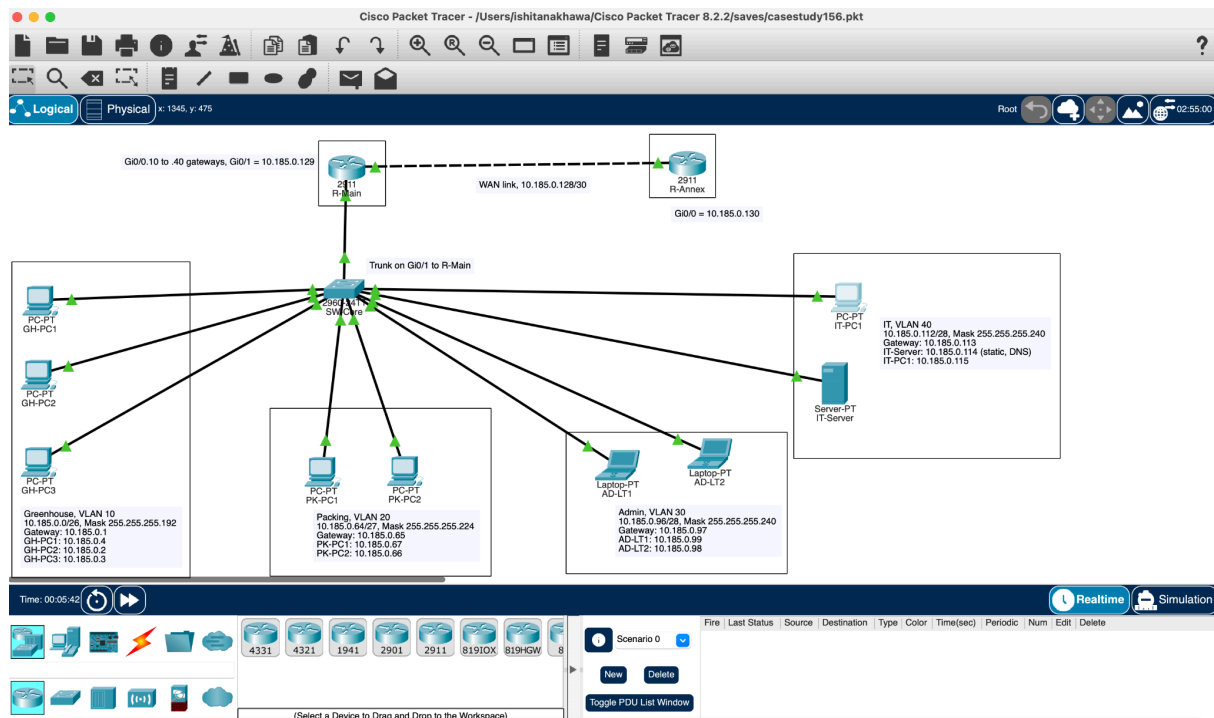
Test	Result
show vlan brief on SW-Core.	Four VLANs with correct ports
show ip interface brief on R-Main.	Four sub-interfaces and WAN port up

DHCP on all 8 clients	Each got an address, mask, gateway, and DNS from its own range.
Ping the gateway from GH-PC1	4 of 4 replies
Ping Packing, Admin and IT hosts	Replies after the first (normal ARP delay)
Ping R-Annex (10.185.0.130)	4 of 4 replies
Ping monitor.canvasback.local	Resolved to 10.185.0.114, 4 of 4 replies

5.4 DHCP, ARP, DNS, and ICMP sequence

When a new climate sensor is plugged into the greenhouse growing floor, it has no address, so it starts with DHCP. It broadcasts a Discover message, R-Main offers an address from the Greenhouse pool together with the gateway 10.185.0.1 and the DNS server 10.185.0.114, the sensor requests that address, and R-Main acknowledges it. Next comes ARP. The monitoring server is on a different subnet, so the sensor must send its traffic to the gateway. It broadcasts "Who has 10.185.0.1?" and the router replies with its MAC address, which the sensor saves. Then DNS: the sensor asks the DNS server to resolve monitor.canvasback.local, and the server answers 10.185.0.114. Finally, ICMP confirms the path works: the sensor sends an Echo Request to 10.185.0.114, the router passes it between VLANs, and the server sends back an Echo Reply. Only after these four steps can the sensor report its readings to the internal monitoring server.

6. Snapshots



Final logical diagram with labels and addresses

```
SW-Core#write memory
Building configuration...
[OK]
SW-Core#show vlan brief

VLAN Name      Status Ports
-----
1  default    active Fa0/20, Fa0/11, Fa0/12, Fa0/13
   Fa0/14, Fa0/15, Fa0/16, Fa0/17
   Fa0/18, Fa0/19, Fa0/20, Fa0/21
   Fa0/22, Fa0/23, Fa0/24, Gi0/1
10  GREENHOUSE active Fa0/7, Fa0/2, Fa0/3
20  PACKING    active Fa0/4, Fa0/5
30  ADMIN      active Fa0/6, Fa0/7
40  IT          active Fa0/8, Fa0/9
1002 token-ring-default active
1003 token-ring-default active
1004 fddienc-default active
1005 trnet-default active
SW-Core#
```

Copy Paste

Top

show vlan brief on SW-Core.

```

R-Main#write memory
Building configuration...
[OK]
R-Main#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	10.185.0.1	YES	manual	up	up
GigabitEthernet0/0.20	10.185.0.65	YES	manual	up	up
GigabitEthernet0/0.30	10.185.0.97	YES	manual	up	up
GigabitEthernet0/0.40	10.185.0.113	YES	manual	up	up
GigabitEthernet0/1	10.185.0.129	YES	manual	up	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

R-Main#

```

Show the IP interface brief on R-Main.

```

R-Main#show ip dhcp pool

```

Pool GREENHOUSE :

Utilization mark (high/low)	: 100 / 0
Subnet size (first/next)	: 0 / 0
Total addresses	: 62
Leased addresses	: 0
Excluded addresses	: 4
Pending event	: none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
10.185.0.1	10.185.0.1 - 10.185.0.62	0 / 4 / 62

Pool PACKING :

Utilization mark (high/low)	: 100 / 0
Subnet size (first/next)	: 0 / 0
Total addresses	: 30
Leased addresses	: 0
Excluded addresses	: 4
Pending event	: none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
10.185.0.65	10.185.0.65 - 10.185.0.94	0 / 4 / 30

Pool ADMIN :

Utilization mark (high/low)	: 100 / 0
Subnet size (first/next)	: 0 / 0
Total addresses	: 14
Leased addresses	: 0
Excluded addresses	: 4
Pending event	: none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
10.185.0.97	10.185.0.97 - 10.185.0.110	0 / 4 / 14

Pool IT :

Utilization mark (high/low)	: 100 / 0
Subnet size (first/next)	: 0 / 0
Total addresses	: 14
Leased addresses	: 0
Excluded addresses	: 4
Pending event	: none

1 subnet is currently in the pool

Current index	IP address range	Leased/Excluded/Total
10.185.0.113	10.185.0.113 - 10.185.0.126	0 / 4 / 14

```

R-Main#

```

show ip dhcp pool on R-Main

```

-----
R-Annex#ping 10.185.0.129

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.185.0.129, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R-Annex#

```

R-Annex interface and successful ping to 10.185.0.129

IT-Server

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

Off

Resource Records

Name

Type

A Record

Address

Add

Save

Remove

No.	Name	Type	Detail
0	monitor.canvasback.local	A Record	10.185.0.114

DNS Cache

Top

IT-Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

DHCP

Static

IPv4 Address

10.185.0.114

Subnet Mask

255.255.255.240

Default Gateway

10.185.0.113

DNS Server

10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::201:97FF:FEAB:E750

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top

IT-Server static IP and DNS record

GH-PC1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.4

Subnet Mask255.255.255.192

Default Gateway10.185.0.1

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address /

Link Local AddressFE80::290:CFF:FE4B:4991

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

GH-PC2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.2

Subnet Mask255.255.255.192

Default Gateway10.185.0.1

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local AddressFE80::206:2AFF:FEB1:69AA

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

GH-PC3

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.3

Subnet Mask255.255.255.192

Default Gateway10.185.0.1

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local AddressFE80::210:11FF:FEAB:1C13

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

PK-PC1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.67

Subnet Mask255.255.255.224

Default Gateway10.185.0.65

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::201:96FF:FE8B:E1C

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

PK-PC2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.66

Subnet Mask255.255.255.224

Default Gateway10.185.0.65

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::20C:85FF:FE73:709

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

AD-LT1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.99

Subnet Mask255.255.255.240

Default Gateway10.185.0.97

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address/

Link Local AddressFE80::230:F2FF:FE35:B272

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

AD-LT2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.98

Subnet Mask255.255.255.240

Default Gateway10.185.0.97

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::20D:BDFF:FE65:91A8

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

Top

IT-PC1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

X

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address10.185.0.115

Subnet Mask255.255.255.240

Default Gateway10.185.0.113

DNS Server10.185.0.114

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local AddressFE80::20D:BDFF:FED9:BB79

Default Gateway

DNS Server

802.1X

Use 802.1X Security

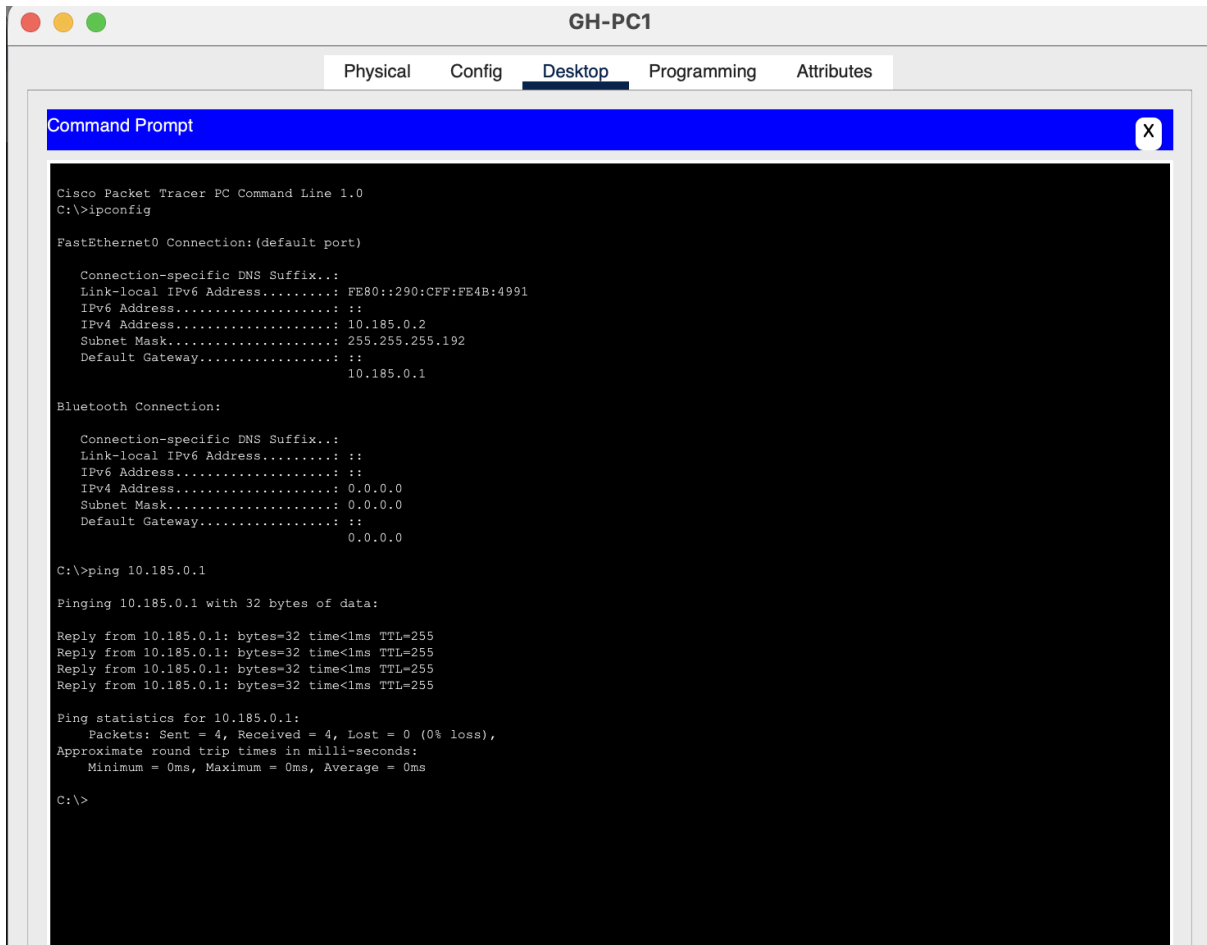
AuthenticationMD5

Username

Password

Top

DHCP result on one PC from each department



```
C:\>ping 10.185.0.66

Pinging 10.185.0.66 with 32 bytes of data:

Request timed out.
Reply from 10.185.0.66: bytes=32 time<1ms TTL=127
Reply from 10.185.0.66: bytes=32 time<1ms TTL=127
Reply from 10.185.0.66: bytes=32 time<1ms TTL=127

Ping statistics for 10.185.0.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.185.0.98

Pinging 10.185.0.98 with 32 bytes of data:

Request timed out.
Reply from 10.185.0.98: bytes=32 time=1ms TTL=127
Reply from 10.185.0.98: bytes=32 time<1ms TTL=127
Reply from 10.185.0.98: bytes=32 time<1ms TTL=127

Ping statistics for 10.185.0.98:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.185.0.115

Pinging 10.185.0.115 with 32 bytes of data:

Request timed out.
Reply from 10.185.0.115: bytes=32 time<1ms TTL=127
Reply from 10.185.0.115: bytes=32 time<1ms TTL=127
Reply from 10.185.0.115: bytes=32 time<1ms TTL=127

Ping statistics for 10.185.0.115:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>ping 10.185.0.130
```

```
Pinging 10.185.0.130 with 32 bytes of data:
```

```
Reply from 10.185.0.130: bytes=32 time<1ms TTL=254
Reply from 10.185.0.130: bytes=32 time<1ms TTL=254
Reply from 10.185.0.130: bytes=32 time<1ms TTL=254
Reply from 10.185.0.130: bytes=32 time<1ms TTL=254
```

```
Ping statistics for 10.185.0.130:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping monitor.canvasback.local
```

```
Pinging 10.185.0.114 with 32 bytes of data:
```

```
Reply from 10.185.0.114: bytes=32 time<1ms TTL=127
Reply from 10.185.0.114: bytes=32 time<1ms TTL=127
Reply from 10.185.0.114: bytes=32 time<1ms TTL=127
Reply from 10.185.0.114: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 10.185.0.114:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>arp -a
```

Internet Address	Physical Address	Type
10.185.0.1	0004.9aa1.3e01	dynamic

```
C:\>
```

```
C:\>arp -d
```

```
C:\>
```

Ping tests, DNS ping, and arp -a

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 1112, y: 52

Root 05:48:00

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	GH-PC1	DHCP
	0.000	--	GH-PC1	DHCP
	0.001	GH-PC1	SW-Core	DHCP
	0.001	--	GH-PC1	DHCP
	0.002	GH-PC1	SW-Core	DHCP
	0.002	SW-Core	GH-PC2	DHCP
	0.002	SW-Core	GH-PC3	DHCP
	0.002	SW-Core	R-Main	DHCP
	0.003	SW-Core	GH-PC2	DHCP
	0.003	SW-Core	GH-PC3	DHCP
	0.003	SW-Core	R-Main	DHCP
	1.511	R-Main	SW-Core	DHCP
	1.512	SW-Core	GH-PC1	DHCP
	1.512	SW-Core	GH-PC2	DHCP
	1.512	SW-Core	GH-PC3	DHCP

Reset Simulation ☒ Constant Delay

Captured to: 1.512 s

Play Controls

Event List Filters - Visible Events
DHCP

Edit Filters Show All/None

Time: 00:16:48.015 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

(Select a Device to Drag and Drop to the Workspace)

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 1112, y: 52

Root 06:07:00

Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.002	SW-Core	R-Main	DHCP
	0.003	SW-Core	GH-PC2	DHCP
	0.003	SW-Core	GH-PC3	DHCP
	0.003	SW-Core	R-Main	DHCP
	1.511	R-Main	SW-Core	DHCP
	1.512	SW-Core	GH-PC1	DHCP
	1.512	SW-Core	GH-PC2	DHCP
	1.512	SW-Core	GH-PC3	DHCP
	1.513	GH-PC1	SW-Core	DHCP
	1.514	SW-Core	GH-PC2	DHCP
	1.514	SW-Core	GH-PC3	DHCP
	1.514	SW-Core	R-Main	DHCP
	1.515	R-Main	SW-Core	DHCP
	1.516	SW-Core	GH-PC1	DHCP
	1.516	SW-Core	GH-PC2	DHCP
	1.516	SW-Core	GH-PC3	DHCP

Reset Simulation ☒ Constant Delay

Captured to: 1.516 s

Play Controls

Event List Filters - Visible Events
DHCP

Edit Filters Show All/None

Time: 00:16:48.015 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

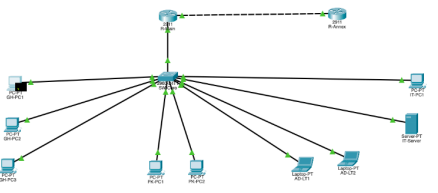
(Select a Device to Drag and Drop to the Workspace)

DHCP exchange (Discover, Offer, Request, ACK)—simulation event list

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 952, y: 740

Root 09:08:30



Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	GH-PC1	ARP
	0.000	--	GH-PC1	DNS
	0.000	--	GH-PC1	ARP
	0.001	GH-PC1	SW-Core	ARP
	0.001	--	GH-PC1	ARP
	0.002	GH-PC1	SW-Core	ARP
	0.002	SW-Core	GH-PC2	ARP
	0.002	SW-Core	GH-PC3	ARP
	0.002	SW-Core	R-Main	ARP
	0.003	SW-Core	GH-PC2	ARP
	0.003	SW-Core	GH-PC3	ARP
	0.003	SW-Core	R-Main	ARP
	0.004	R-Main	SW-Core	ARP
	0.005	SW-Core	GH-PC1	ARP
	0.005	--	GH-PC1	DNS
	0.006	GH-PC1	SW-Core	DNS

Reset Simulation Constant Delay Captured to: 0.021 s

Play Controls

Event List Filters - Visible Events
ARP, DNS, ICMP

Edit Filters Show All/None

Time: 00:16:58.028 PLAY CONTROLS: [Stop] [Play] [Pause] [Fast Forward]

Scenario 0 [New] [Delete] [Toggle PDU List Window]

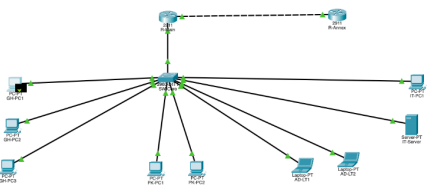
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete

(Select a Device to Drag and Drop to the Workspace)

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 952, y: 740

Root 09:12:30



Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.006	GH-PC1	SW-Core	DNS
	0.007	SW-Core	R-Main	DNS
	0.008	R-Main	SW-Core	DNS
	0.009	SW-Core	IT-Server	DNS
	0.009	SW-Core	IT-PC1	DNS
	0.010	IT-Server	SW-Core	DNS
	0.011	SW-Core	R-Main	DNS
	0.012	R-Main	SW-Core	DNS
	0.013	SW-Core	GH-PC1	DNS
	0.013	--	GH-PC1	ICMP
	0.014	GH-PC1	SW-Core	ICMP
	0.015	SW-Core	R-Main	ICMP
	0.016	R-Main	SW-Core	ICMP
	0.017	SW-Core	IT-Server	ICMP
	0.018	IT-Server	SW-Core	ICMP
	0.019	SW-Core	R-Main	ICMP

Reset Simulation Constant Delay Captured to: 0.021 s

Play Controls

Event List Filters - Visible Events
ARP, DNS, ICMP

Edit Filters Show All/None

Time: 00:16:58.028 PLAY CONTROLS: [Stop] [Play] [Pause] [Fast Forward]

Scenario 0 [New] [Delete] [Toggle PDU List Window]

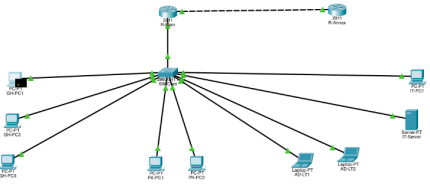
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete

(Select a Device to Drag and Drop to the Workspace)

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 952, y: 740

Root 09:19:00



Simulation Panel

Vis.	Time(sec)	Last Device	At Device	Type
	0.008	H-Main	SW-Core	DNS
	0.009	SW-Core	IT-Server	DNS
	0.009	SW-Core	IT-PC1	DNS
	0.010	IT-Server	SW-Core	DNS
	0.011	SW-Core	R-Main	DNS
	0.012	R-Main	SW-Core	DNS
	0.013	SW-Core	GH-PC1	DNS
	0.013	--	GH-PC1	ICMP
	0.014	GH-PC1	SW-Core	ICMP
	0.015	SW-Core	R-Main	ICMP
	0.016	R-Main	SW-Core	ICMP
	0.017	SW-Core	IT-Server	ICMP
	0.018	IT-Server	SW-Core	ICMP
	0.019	SW-Core	R-Main	ICMP
	0.020	R-Main	SW-Core	ICMP
	0.021	SW-Core	GH-PC1	ICMP

Reset Simulation ☒ Constant Delay Captured to: 0.021 s

Play Controls

Event List Filters - Visible Events
ARP, DNS, ICMP

Edit Filters Show All/None

Time: 00:16:58.028 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

(Select a Device to Drag and Drop to the Workspace)

ARP, DNS, and ICMP sequence Simulation Event List

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 1078, y: 510

Root 10:19:00

PDU Information at Device: IT-Server

OST Model Inbound PDU Details Outbound PDU Details

At Device: IT-Server
Source: GH-PC1
Destination: 10.185.0.114

In Layers

Layer 7: DNS

Layer 6

Layer 5

Layer 4: UDP Src Port: 1026, Dst Port: 53

Layer 3: IP Header Src. IP: 10.185.0.4, Dest. IP: 10.185.0.114

Layer 2: Ethernet II Header 0004.9AA1.3E01 >> 0001.97AB.E750

Layer 1: Port FastEthernet0

Out Layers

Layer 7: DNS

Layer 6

Layer 5

Layer 4: UDP Src Port: 53, Dst Port: 1026

Layer 3: IP Header Src. IP: 10.185.0.114, Dest. IP: 10.185.0.4

Layer 2: Ethernet II Header 0001.97AB.E750 >> 0004.9AA1.3E01

Layer 1: Port(s): FastEthernet0

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

Simulation Panel

At Device	Type
SW-Core	UNIS
IT-Server	DNS
IT-PC1	DNS
SW-Core	DNS
R-Main	DNS
SW-Core	DNS
GH-PC1	DNS
SW-Core	ICMP
R-Main	ICMP
SW-Core	ICMP
IT-Server	ICMP
SW-Core	ICMP
R-Main	ICMP
SW-Core	ICMP
GH-PC1	ICMP

Captured to: 0.021 s

Time: 00:16:58.028 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

(Select a Device to Drag and Drop to the Workspace)

Cisco Packet Tracer - /Users/ishtanakhawa/Cisco Packet Tracer 8.2.2/saves/casestudy156.pkt

Logical Physical x: 1146, y: 574

Root 10:26:00

PDU Information at Device: IT-Server

OST Model Inbound PDU Details Outbound PDU Details

At Device: IT-Server
Source: GH-PC1
Destination: 10.185.0.114

In Layers

Layer 7

Layer 6

Layer 5

Layer 4

Layer 3: IP Header Src. IP: 10.185.0.4, Dest. IP: 10.185.0.114 ICMP Message Type: 8

Layer 2: Ethernet II Header 0004.9AA1.3E01 >> 0001.97AB.E750

Layer 1: Port FastEthernet0

Out Layers

Layer 7

Layer 6

Layer 5

Layer 4

Layer 3: IP Header Src. IP: 10.185.0.114, Dest. IP: 10.185.0.4 ICMP Message Type: 0

Layer 2: Ethernet II Header 0001.97AB.E750 >> 0004.9AA1.3E01

Layer 1: Port(s): FastEthernet0

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

Simulation Panel

At Device	Type
SW-Core	UNIS
IT-Server	DNS
IT-PC1	DNS
SW-Core	DNS
R-Main	DNS
SW-Core	DNS
GH-PC1	DNS
SW-Core	ICMP
GH-PC1	ICMP
SW-Core	ICMP
R-Main	ICMP
SW-Core	ICMP
IT-Server	ICMP
SW-Core	ICMP
R-Main	ICMP
SW-Core	ICMP
GH-PC1	ICMP

Captured to: 0.021 s

Time: 00:16:58.028 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

(Select a Device to Drag and Drop to the Workspace)

OSI Model tab for a DNS and an ICMP packet

7. Conclusion

The project produced a complete, working network design for Canvasback Terrarium & Houseplant Subscription Co. An extended star topology was chosen for its low cost, good fault isolation, and easy growth. VLSM divided 10.185.0.0/22 into five efficient blocks with 20% spare capacity, leaving the rest of the range free for the future. Four VLANs keep each department's traffic separate, one router connects them, and a private point-to-point link reaches the Annex. DHCP and DNS were set up so devices configure themselves and can be found by name. Tests in Cisco Packet Tracer confirmed that every department can reach the others, the Annex, and the DNS server, and the captured packets showed the DHCP, ARP, DNS, and ICMP sequence in the expected order. Possible improvements are a second core switch for redundancy, access points for wireless sensors, a firewall, and NAT for internet access.

8. Reference

1. Cisco Packet Tracer documentation, Cisco Networking Academy.
2. RFC 1918, Address Allocation for Private Internets.
3. RFC 4632, Classless Inter-domain Routing (CIDR).
4. RFC 2131, Dynamic Host Configuration Protocol.
5. RFC 826, An Ethernet Address Resolution Protocol.
6. RFC 1034 and RFC 1035, Domain Names: Concepts and Implementation.
7. RFC 792, Internet Control Message Protocol.
8. IEEE 802.1Q, Virtual Bridged Local Area Networks.
9. Course material: Computer Networking & Cyber Security, B.Tech CSE 2025-29, Case Study no. 156.